

Molecular representations, similarity and molecular descriptors

Molecular representation –why it matters

- **Molecules as data**

- Computers need structured data
- Molecular representations translate chemistry into numbers, graphs and strings
- Algorithms can reason about chemical space computationally

- **2D- versus 3D representations**

- 2D – connectivity and bonding (SMILES, InChI)
- 3D – spatial geometry, conformation, chirality (pdb, sdf, mol2)
- Drug activity often depends **shape** and **orientation** – not just formula

- **Standardization and interoperability**

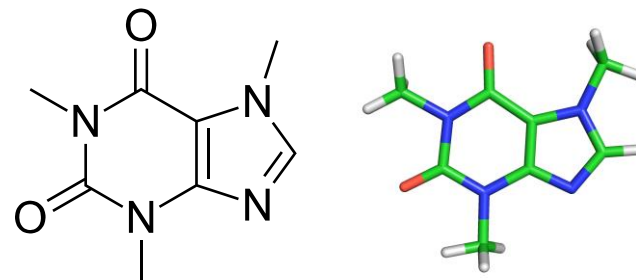
- Formats (SDF, MOL2, PDB, SMILES) ensure compatibility across software
- Reproducibility depends on **consistent molecular definitions** (e.g. tautomers, protonation states)
- Critical for automated workflows like virtual screening and QSAR

- **From representation to prediction**

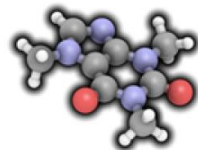
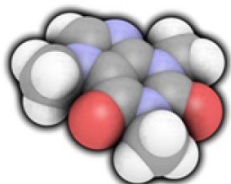
- The quality of the model depends on how well it represents reality
- Proper encoding needed for **docking, scoring, and ML-based property prediction**
- “Garbage in, garbage out” — accurate representations are the foundation of reliable results.

Representing chemical structures

- Exampe: representing the molecule of caffeine in different ways



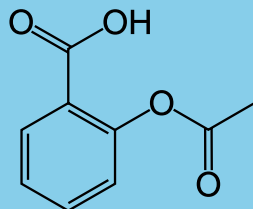
Representation Name	Representation of Caffeine
Common Name	Caffeine
Synonyms	Guaranine
	Methyltheobromine
	1,3,7-Trimethylxanthine
	Theine
Empirical Formula	C ₈ H ₁₀ N ₄ O ₂
IUPAC Name	1,3,7-trimethylpurine-2,6-dione
CAS Registry Number	58-08-2
ChEMBL ID	CHEMBL113
Wiswesser Line Notation (WLN)	T56 BN DN FNVNVJ B F H
SMILES	<chem>CN1C=NC2=C1C(=O)N(C(=O)N2C)C</chem>
Aromatic SMILES	<chem>CN1C(=O)N(C)c2ncn(C)2C1=O</chem>

Representation Name	Representation of Caffeine
InChI	1S/C8H10N4O2/c1-10-4-9-6- 5(10)7(13)12(3)8(14)11(6)2/h4H,1-3H3
InChIKey	RYYVLZVUVIJVGH-UHFFFAOYSA-N
Topography	
Surface	

2D molecular representation

$C_9H_8O_4$

O=C(C)Oc1ccccc1C(=O)O



InChI=1S/C9H8O4/c1-6(10)13-8-5-3-2-4-7(8)9(11)12/h2-5H,1H3,(H,11,12)

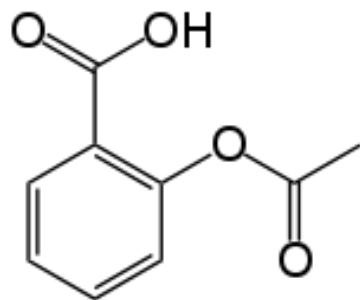
Key:BSYNRYMUTXBXSQ-UHFFFAOYSA-N

- **Abstract the molecule's topology**
 - Capture how atoms are connected through bonds, without explicit 3D coordinates
- **Serve as the foundation for cheminformatics**
 - Used in databases, QSAR models and similarity searches
- **Common formats**
 - SMILES, InChI, and connection tables (as in MOL or SDF files).
- **Efficient for computation**
 - enable fast storage, indexing, and comparison of large compound libraries.
- **Limitations**
 - 2D formats cannot capture spatial arrangements, conformations and geometry information like bond lengths, bond angles, etc..

SMILES (1)

SMILES - Simplified Input Molecular Entry Specification

Linguagem que permite a representação de estruturas moleculares 2D na forma de uma sequência (“string”) de caracteres.



Estrutura 2D



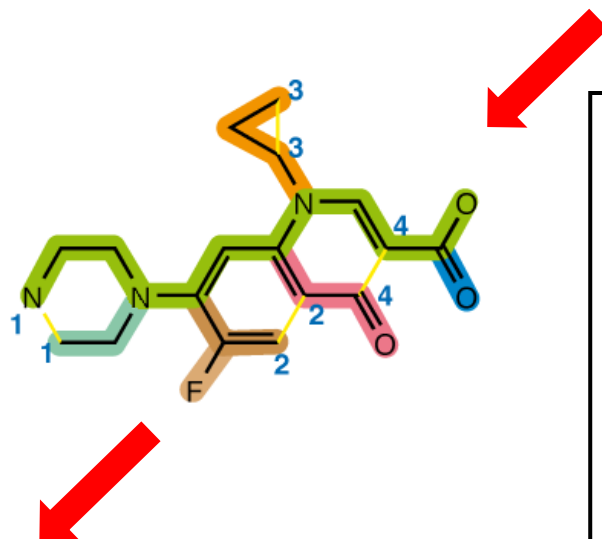
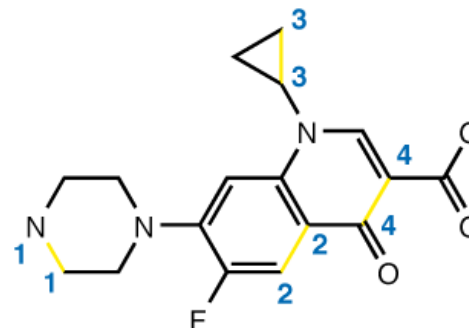
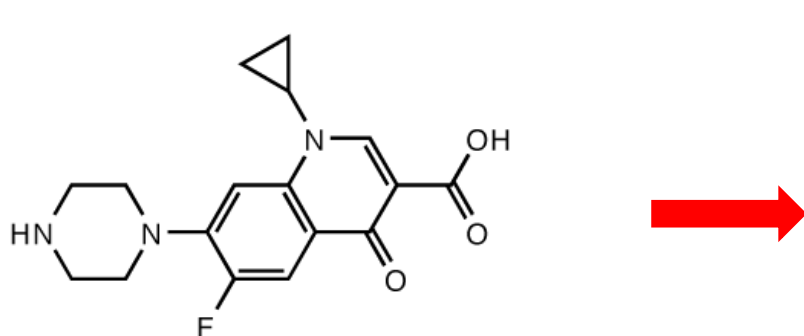
O=C(Oc1ccccc1C(=O)O)C

SMILES

Tutorial SMILES: <http://www.daylight.com/>

D. Weininger (1988) *J. Chem. Inf. Comput. Sci.* **28**:31

SMILES (2)

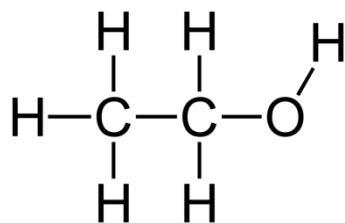


1. Chose the longest bond path
2. All other paths are branches ()
3. Rings are open, and atoms by cut bonds numbered
4. Double bonds are represent by "=", single bonds implicit
5. Hydrogens are (generally) implict

N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O

Canonical SMILES

- SMILES strings are not uniquely defined for each molecule, for example:



ethanol

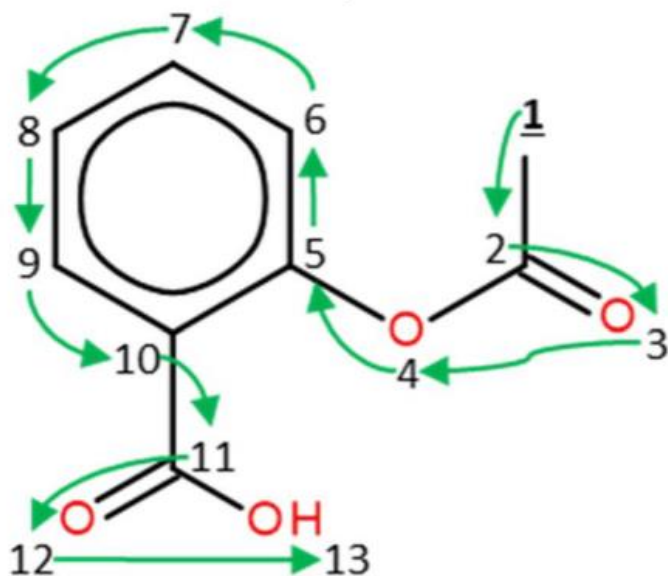


All valid SMILES representations

- To solve this problem, *canonization algorithms* that produce a unique SMILES string for each structure have been developed
- Many software packages can generate canonical SMILES:
- Daylight Chemical Information Systems
- OpenEye Scientific Software
- Chemical Computing Group
- Chemistry Development Kit

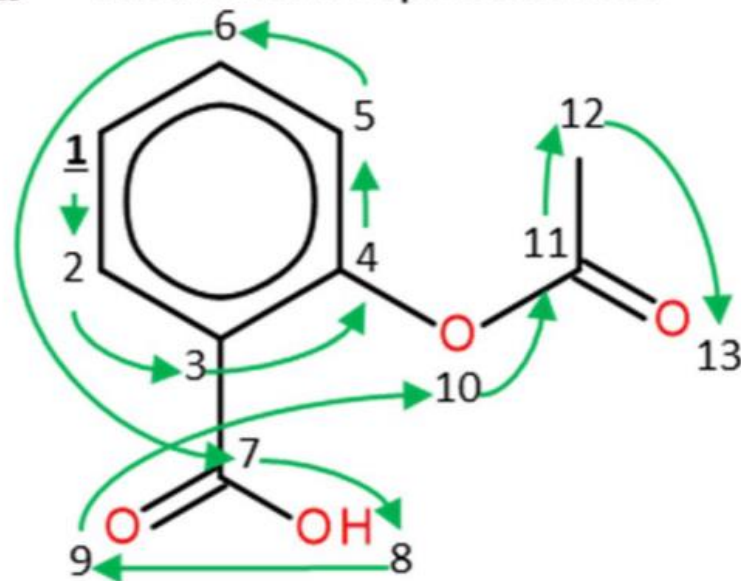
Canonical versus randomized SMILES

a Canonical representation



CC(=O)Oc1ccccc1C(=O)O

b Randomized representation



c1cc(c(cc1)C(O)=O)OC(C)=O

InChI representation

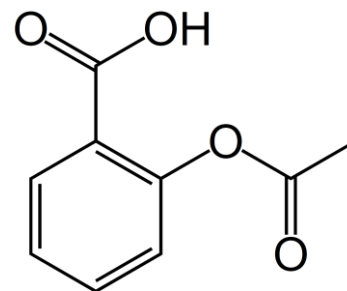
InChI – IUPAC International Chemical Identifier

Developed by IUPAC and NIST 2000-2005 (pronounced “In Key”)

InChI is a text-based identifier for chemical substances, designed to offer a standard way to provide molecular information

The InChI Identifier describes molecules in terms of different layers of information:

- Main Layer
 - Chemical Formula
 - Atom connections
 - Hydrogen Atoms
- Charge Layer
- Stereochemical Layer
- Isotopic Layer
- Fixed-H Layer
- Reconnected layer



InChI=1S/C9H8O4/c1-6(10)13-8-5-3-2-4-7(8)9(11)12/h2-5H,1H3,(H,11,12)

1-version number
S-standardized InChI

**Chemical
Formula**

Connectivity

**Hydrogen
Atoms**

InChI - Rifampicin

InChI=1S/C43H58N4O12/c1-21-12-11-13-22(2)42(55)45-33-28(20-44-47-17-15-46(9)16-18-47)37(52)30-31(38(33)53)36(51)26(6)40-32(30)41(54)43(8,59-40)57-19-14-29(56-10)23(3)39(58-27(7)48)25(5)35(50)24(4)34(21)49/h11-14,19-21,23-25,29,34-35,39,49-53H,15-18H2,1-10H3,(H,45,55)/b12-11+,19-14+,22-13-,44-20-/t21-,23+,24+,25+,29-,34-,35+,39+,43-/m0/s1

Chemical Formula - C₄₃H₅₈N₄O₁₂

Connectivity - c1-21-12-11-13...

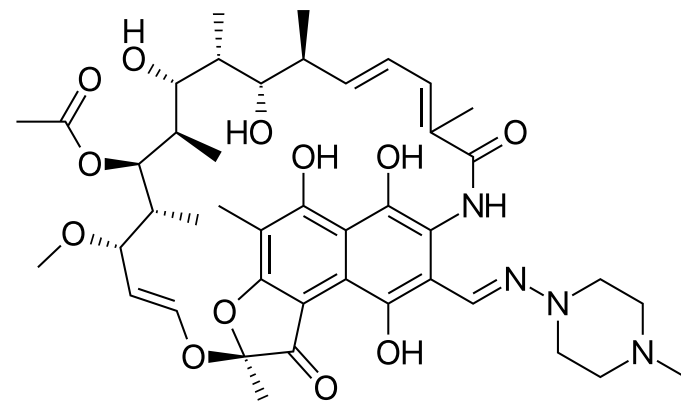
Hydrogen atoms - h11-14,19-21,23-25,29,3...

Double bond stereochemistry - b12-11+,19-14+,22-13-...

Tetrahedral stereochemistry - t21-,23+,24+,25+,29-,34-...

m0 – absolute stereochemical configuration

S1 – no inversion



- Ansamycin antibiotic with complex structure
- 9 chiral centers
- 4 double bonds with E/Z configurations
- Hetero-ring systems

InChI is canonical by default

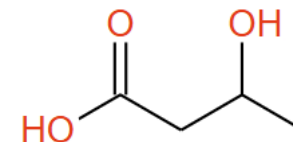
- For InChI generation, atoms are numbered according to a canonical algorithm that ensures unique sequence for a given structures

Same molecule:

CC(O)CC(=O)O InChI=1S/C4H8O3/c1-3(5)2-4(6)7/h3,5H,2H2,1H3,(H,6,7)

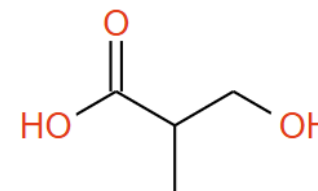
C(C(C)O)C(=O)O InChI=1S/C4H8O3/c1-3(5)2-4(6)7/h3,5H,2H2,1H3,(H,6,7)

C(CC(=O)O)(C)O InChI=1S/C4H8O3/c1-3(5)2-4(6)7/h3,5H,2H2,1H3,(H,6,7)



Different molecule:

OCC(C)C(=O)O InChI=1S/C4H8O3/c1-3(2-5)4(6)7/h3,5H,2H2,1H3,(H,6,7)



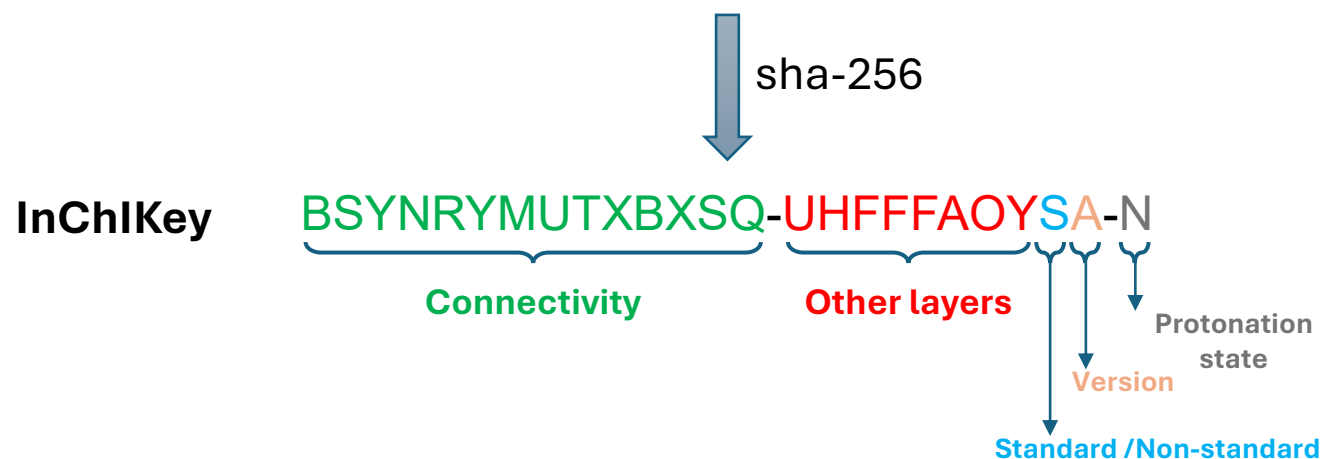
InChI key

InChI's are too long and complex to reliably work as search keywords in database/internet searches.

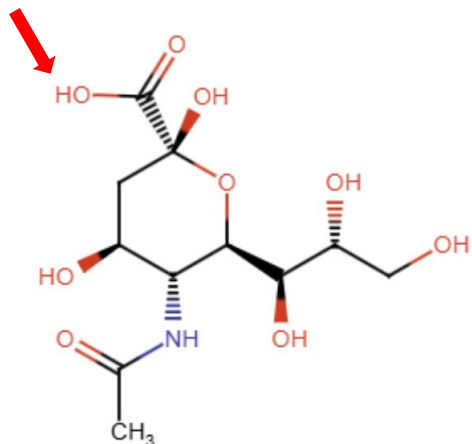
InChI Key – Compressed form of the InChI, using a hashing algorithm (sha-256) to produce a quasi-unique alphabetic string with shorter length.

Different InChI's can produce the same InChI Key, but that's an extremely rare event. The **connectivity** layer is hashed separately – two molecules with the same connectivity but with different spatial arrangement will have the same first part of InChI Key

InChI=1S/C9H8O4/**c**1-6(10)13-8-5-3-2-4-7(8)9(11)12/**h**2-5H,1H3,(H,11,12)

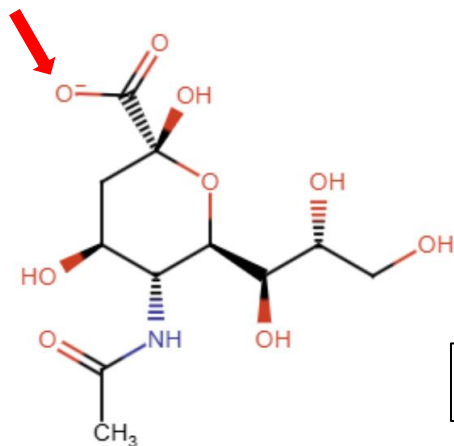


InChI key – sialic acid



N-acetyl-alpha-neuraminic acid

SQVRNKJHWKZAKO-YRMXFSIDSA-**N**

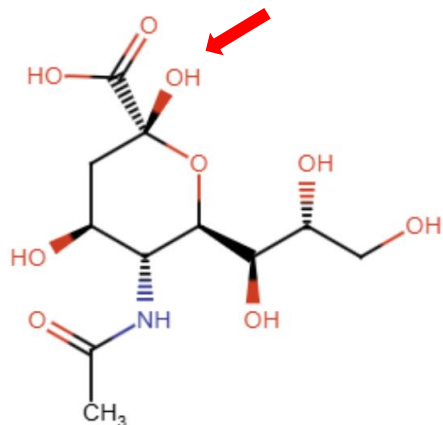


N-acetyl-alpha-neuramate

SQVRNKJHWKZAKO-YRMXFSIDSA-**M**

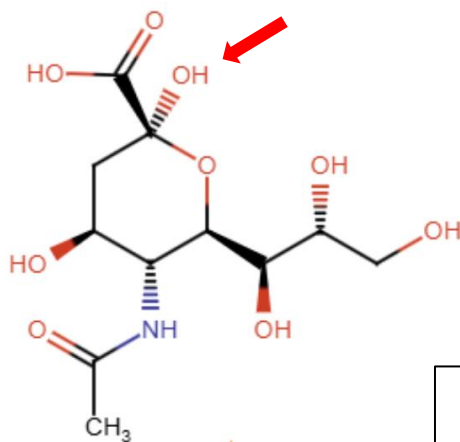
Change is only in **protonation state**

InChI key – sialic acid



N-acetyl-alpha-neuraminic acid

SQVRNKJHWKZAKO-**YRMXFSIDSA**-N



N-acetyl-beta-neuraminic acid

SQVRNKJHWKZAKO-**PFQGKNLYSA**-N

Change is only in stereo-isomerism layer of InChI

Online InChI(Key) generator



InChI Web Demo

<https://iupac-inchi.github.io/InChI-Web-Demo/>

InChI RInChI About

Draw structure and convert to InChI Convert Molfile to InChI Convert Auxinfo to structure Convert SDF to InChIs

Version: Latest
<https://github.com/IUPAC-InChI/InChI/releases/tag/v1.07.4>

Options:
Tautomer options
☒ Mobile H Perception
☐ Do not add implicit H atoms
☒ Include Stereo: ☒ Absolute
☐ Relative
☐ Racemic
☐ From chiral flag
☐ Always include omitted/undefined stereo
☐ Different marks for unknown/undefined stereo
☐ Both ends of wedge point to stereocenters
☐ Include Bonds to Metal
☐ Treat polymers:
☐ No pre-edits of original polymer structure
☐ Enable CRU folding
☐ Disable CRU frame shift
☐ Allow non-polymer Zz pseudoatoms
[Reset InChI Options](#)

InChI
InChI=1S/C43H58N4O12/c1-21-12-11-13-22(2)42(55)45-33-28(20-44-47-17-15-46(9)16-18-47)37(52)30-31(38(33)53)36(51)26(6)40-32(30)41(54)43(8,59-40)57-19-14-29(56-10)23(3)39(58-27(7)48)25(5)35(50)24(4)34(21)49/h11-14,19-21,23-25,29,34-35,39,49-53H,15-18H2,1-10H3,(H,45,55)/b12-11+,19-14+,22-13-,44-20+/t21-,23+,24+,25+,29-,34-,35+,39+,43-/m0/s1

InChIKey
JQXXHWPUNPDR-TLSIYKJHSA-N

The importance of 3D representations

- **Capture the true shape of molecules:** 3D coordinates define geometry, orientation, and chirality, essential for biological activity.
- **Enable structure-based methods:** Docking, molecular dynamics, and pharmacophore modelling all rely on accurate 3D structures.
- **Bridge experimental and computational worlds:** Derived from crystallography (PDB, CIF) or built computationally (RDKit, Open Babel).
- **Determine molecular interactions:** 3D structure governs binding affinity, selectivity, and conformational flexibility.

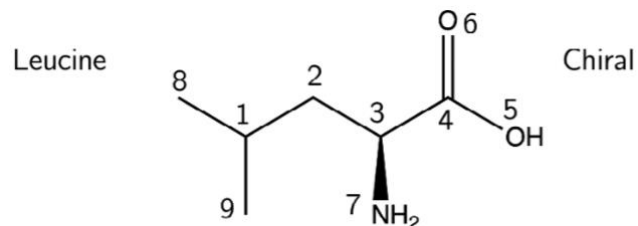
Common 3D representation formats

- **PDB (.pdb) :**
 - Standard format for macromolecules (proteins, nucleic acids, complexes); includes atomic coordinates and metadata such as chains, residues, and secondary structure.
- **MOL2 (.mol2) :**
 - Tripos format supporting atom types, charges, and bonding information; widely used for docking and molecular mechanics.
- **SDF (.sdf) :**
 - Structure Data File; can store multiple molecules with associated properties (e.g., activity data, ID numbers).
- **XYZ (.xyz) :**
 - Simple Cartesian coordinate format; easy to read, lacks bonding information.
- **CIF (.cif) :**
 - Crystallographic Information File; used for crystal structures and symmetry information (mainly from X-ray data).
- **PQR (.pqr) :**
 - Derived from PDB, includes partial atomic charges and radii (for electrostatics calculations like APBS)
- **PDBQT(.pdbqt):**
 - Derived from PDB, includes atom types, charges and free torsions (used by docking programs like Autodock and Autodock Vina)

Structure format comparison

Format	Meaning / Origin	Typical Use	Key Features	Notes / Limitations
MOL (.mol)	MDL Molfile	Small molecules, editors like ChemDraw or RDKit	Single molecule; atoms, bonds, 2D/3D coordinates	No metadata or multiple structures
SDF (.sdf)	Structure Data File (MDL)	Compound libraries, QSAR datasets	Multiple MOL blocks + properties (activity, ID, etc.)	No atom types or charges; not ideal for simulations
MOL2 (.mol2)	Tripos Mol2 (SYBYL)	Docking, molecular mechanics	Atom types, charges, substructures, residues	More detailed, but less standardized across tools
PDB (.pdb)	Protein Data Bank format	Proteins, nucleic acids, large complexes	3D atomic coordinates, chains, residues, occupancy	Legacy format, limited precision (fixed columns)
CIF (.cif)	Crystallographic Information File	X-ray / neutron crystal structures	Unit cell, symmetry, atomic positions, metadata	Text-based, rich but complex (mmCIF used for macromolecules)

MDL mol file



```

9  8  0  0  1  0  0  0  0  0999 V2000
-1.0717 -0.2062 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-0.3572 0.2062 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0.3572 -0.2062 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1.0717 0.2062 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1.7861 -0.2062 0.0000 O 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1.0717 1.0312 0.0000 O 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0.3572 -1.0312 0.0000 N 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-1.7861 0.2062 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-1.0717 -1.0312 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1  2  1  0
2  3  1  0
3  4  1  0
4  5  1  0
4  6  2  0
3  7  1  1
1  8  1  0
1  9  1  0
M  END

```

Counts Line

Atom Block

Bond Block

M END

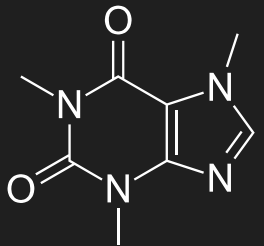
Blocks not used in this Ctab:
Atom List, Stext, Properties

Trypos mol2 format

@<TRIPOS>MOLECULE
 CAFINE
 24 25 1 0 0
 SMALL
 NO_CHARGES

 Generated from the CSD

Chemical structure of Caffeine:



@<TRIPOS>ATOM

Atom	X	Y	Z	Atom types	Charge	Weight	Residue	Occupancy
1 C1	3.6210	3.7158	-0.3933	C.2	1	RES1	0.0000	
2 C2	1.4218	4.2301	0.5103	C.2	1	RES1	0.0000	
3 C3	1.1506	2.9375	0.7660	C.2	1	RES1	0.0000	
4 C4	2.1094	1.9088	0.4551	C.2	1	RES1	0.0000	
5 C5	-0.4705	4.2084	1.4335	C.2	1	RES1	0.0000	
6 C6	4.3372	1.3894	-0.4768	C.3	1	RES1	0.0000	
7 C7	2.9376	6.0755	-0.3117	C.3	1	RES1	0.0000	
8 C8	-0.9085	1.7585	1.8063	C.3	1	RES1	0.0000	
9 N1	3.2629	2.3631	-0.1044	N.am	1	RES1	0.0000	
10 N2	2.6728	4.6242	-0.0599	N.am	1	RES1	0.0000	
11 N3	-0.1337	2.9208	1.3303	N.pl3	1	RES1	0.0000	
12 N4	0.4784	5.0234	0.9615	N.2	1	RES1	0.0000	
13 O1	4.6487	4.0080	-0.9402	O.2	1	RES1	0.0000	
14 O2	1.9391	0.6747	0.6368	O.2	1	RES1	0.0000	
15 H1	-1.5169	4.3587	1.8678	H	1	RES1	0.0000	
16 H2	-0.4822	1.0354	2.3603	H	1	RES1	0.0000	
17 H3	-1.0965	1.0521	1.0954	H	1	RES1	0.0000	
18 H4	-1.8007	2.2879	2.0096	H	1	RES1	0.0000	
19 H5	3.9616	6.0454	-0.5635	H	1	RES1	0.0000	
20 H6	3.3236	6.6132	0.4137	H	1	RES1	0.0000	
21 H7	2.2066	6.2959	-0.8551	H	1	RES1	0.0000	
22 H8	5.2607	1.6700	-0.8984	H	1	RES1	0.0000	
23 H9	4.4294	0.5511	0.0867	H	1	RES1	0.0000	
24 H10	3.9604	1.0020	-1.2767	H	1	RES1	0.0000	

atom names X Y Z atom types
 coordinates

- 3D representation including atoms, charges, atom types, bonds, and other information (.mol2 extension)
- Can be read by many software applications and libraries

bond types

<TRIPOS>BOND

1	1	9	am
2	2	3	2
3	3	4	1
4	4	9	am
5	5	11	1
6	6	9	1
7	7	10	1
8	8	11	1
9	10	1	am
10	11	3	1
11	12	2	1
12	13	1	2
13	14	4	2
14	15	5	1
15	16	8	1
16	17	8	1
17	18	8	1
18	19	7	1
19	20	7	1
20	21	7	1
21	22	6	1
22	23	6	1
23	24	6	1
24	2	10	1
25	5	12	2

<TRIPOS>CRYSTIN

14.8000	16.7000	3.9700	90.0000	97.0000	90.0000	14	2
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From 2D to 3D structure

3D structures may be obtained by different methods

- Experimental: X-ray crystallography, NMR, EPR, Cryo-EM and others
- Theoretical:
 - Quantum Chemistry – accurate, but slow. Limited conformation.
 - From 2D to 3D:
 - **Start with 2D input** — chemical drawings or SMILES encode atomic connectivity but lack spatial information
 - **Generate 3D conformers** — algorithms (e.g., RDKit ETKDG, Open Babel) build plausible 3D geometries from 2D graphs.
 - **Optimize geometry** — force fields (MMFF94, UFF) or quantum methods refine bond lengths and angles to realistic minima.
 - **Validate and export** — save optimized structures in 3D formats (SDF, MOL2, PDB) for use in docking or molecular dynamics.

The importance of molecular similarity

- **Concept and Motivation**

- Similar molecules tend to have similar biological activity — the “similarity principle.”
- Central idea behind **ligand-based drug design**, **virtual screening**, and **QSAR**.
- Helps find new hits when a target’s structure is unknown.

- **How to Measure Similarity**

- Compare **molecular fingerprints** or **descriptors** numerically.
- Common metric: **Tanimoto coefficient** (0 = different, 1 = identical).
- Alternative metrics: Dice, Cosine, Euclidean (for continuous descriptors).

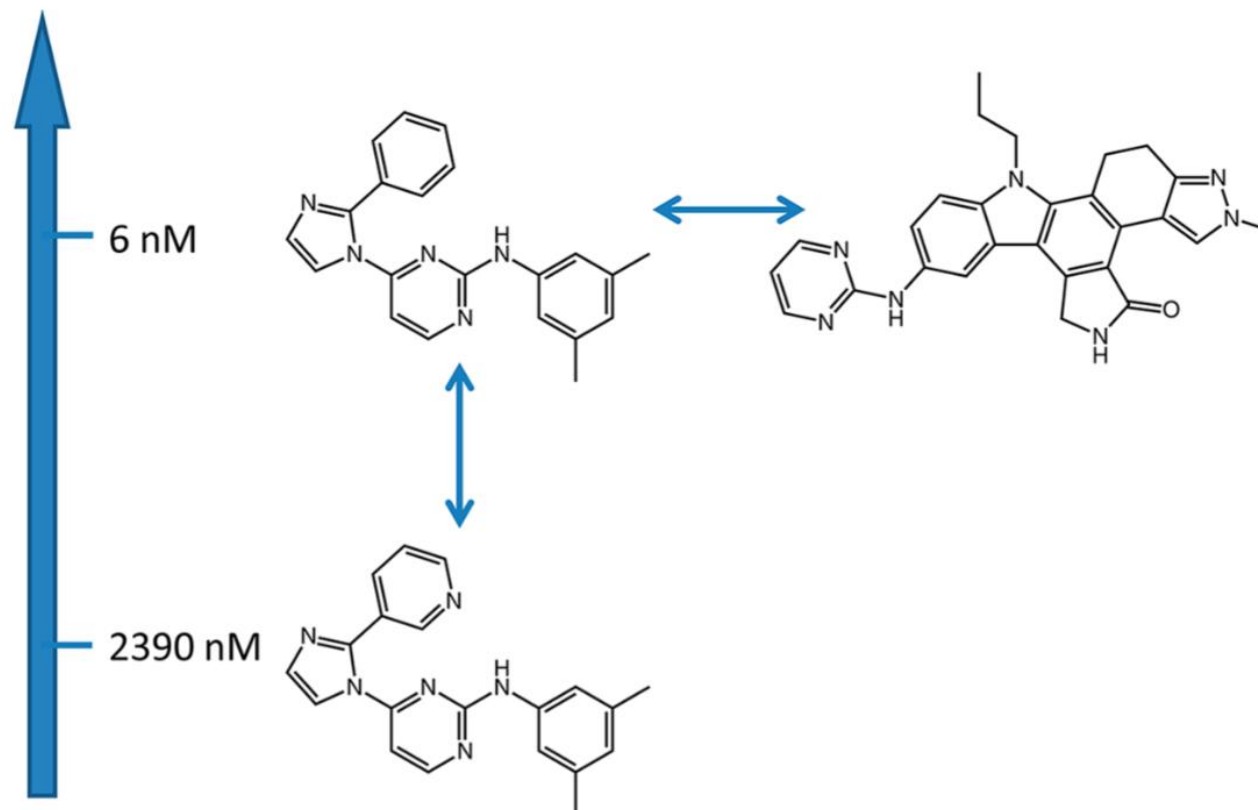
- **Role of Fingerprints and Features**

- Fingerprints capture structural fragments or atom environments (e.g., ECFP, MACCS).
- Descriptors encode physicochemical properties (e.g., logP, MW, H-bond donors).
- The chosen representation defines what “similar” means in a given context.

- **Applications in Drug Design**

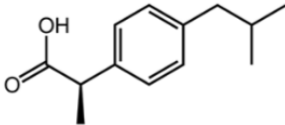
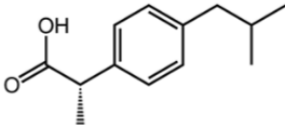
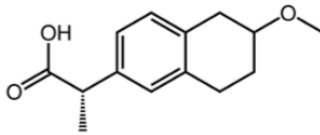
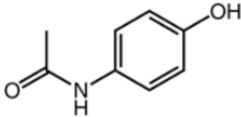
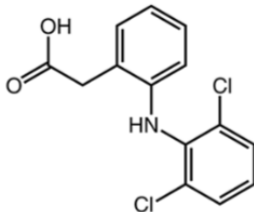
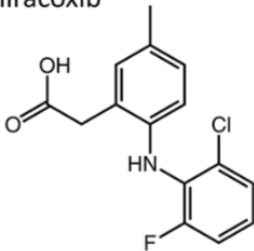
- **Virtual screening**: find analogues of known actives.
- **Scaffold hopping**: identify novel chemotypes with similar pharmacophoric patterns.
- **Activity cliffs**: small structural changes → large activity differences.
- Key tool for prioritizing compounds in early discovery stages.

Similarity versus activity



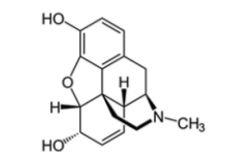
Similarity versus activity. Three vascular endothelial growth factor receptor 2 ligands are shown that represent different (vertical vs horizontal) similarity–activity (potency) relationships.

Complex similarity relationships

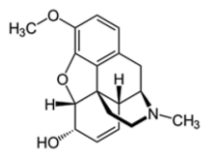
(R)-(-)-Ibuprofen  <table><tr><td>COX-1</td><td>COX-2</td><td>HSL</td></tr><tr><td>no</td><td>no</td><td>no</td></tr></table>	COX-1	COX-2	HSL	no	no	no	(S)-(+)-Ibuprofen  Bioavailability 49 -73% <table><tr><td>COX-1</td><td>COX-2</td><td>HSL</td></tr><tr><td>yes</td><td>yes</td><td>no</td></tr></table>	COX-1	COX-2	HSL	yes	yes	no	(S)-(+)-Naproxen  Bioavailability 95% <table><tr><td>COX-1</td><td>COX-2</td><td>HSL</td></tr><tr><td>yes</td><td>yes</td><td>yes</td></tr></table>	COX-1	COX-2	HSL	yes	yes	yes
COX-1	COX-2	HSL																		
no	no	no																		
COX-1	COX-2	HSL																		
yes	yes	no																		
COX-1	COX-2	HSL																		
yes	yes	yes																		
Paracetamol  Bioavailability ~100% <table><tr><td>COX-1</td><td>COX-2</td><td>HSL</td></tr><tr><td>no</td><td>yes</td><td>no</td></tr></table>	COX-1	COX-2	HSL	no	yes	no	Diclofenac  Bioavailability >99% <table><tr><td>COX-1</td><td>COX-2</td><td>HSL</td></tr><tr><td>yes</td><td>yes</td><td>no</td></tr></table>	COX-1	COX-2	HSL	yes	yes	no	Lumiracoxib  Bioavailability ~74% <table><tr><td>COX-1</td><td>COX-2</td><td>HSL</td></tr><tr><td>no</td><td>yes</td><td>no</td></tr></table>	COX-1	COX-2	HSL	no	yes	no
COX-1	COX-2	HSL																		
no	yes	no																		
COX-1	COX-2	HSL																		
yes	yes	no																		
COX-1	COX-2	HSL																		
no	yes	no																		

Complex similarity relationships. Cyclooxygenase (COX) inhibitors and their activity profiles are compared. HSL – hormone sensitive lipase.

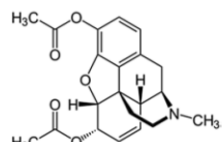
Measuring molecular similarity



morphine

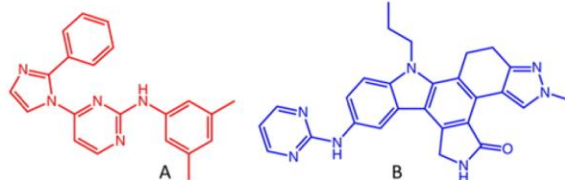
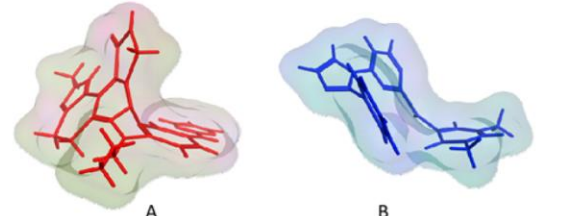
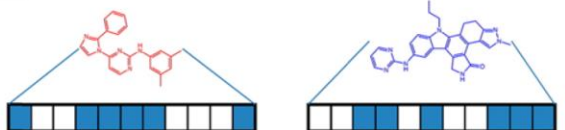
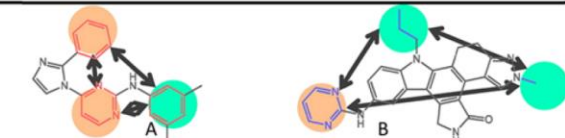


codeine



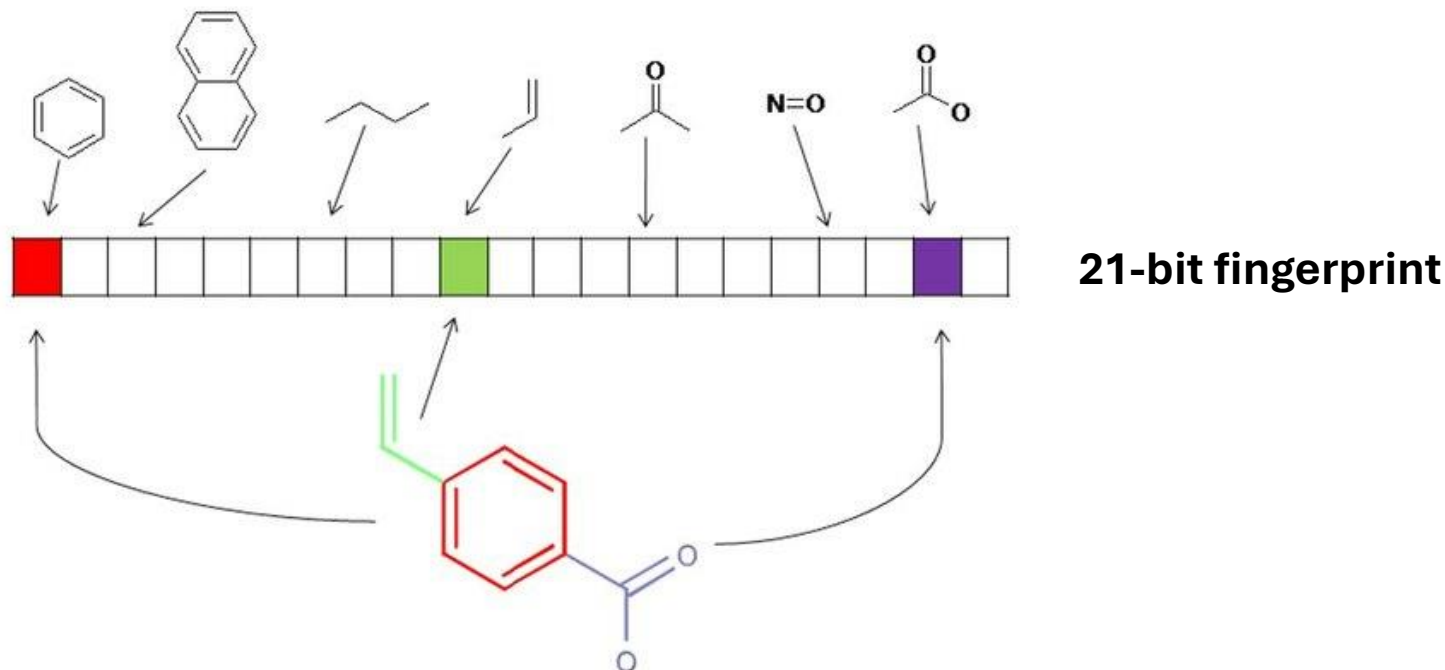
heroin

**Similar structures
similar functions**

Chemical similarity	<table><tr><td></td><td>Mol. weight</td><td>LogP</td><td>Rotatable bonds</td><td>Aromatic rings</td><td>Heavy atoms</td></tr><tr><td>A</td><td>341.4</td><td>5.23</td><td>4</td><td>4</td><td>26</td></tr><tr><td>B</td><td>463.5</td><td>4.43</td><td>4</td><td>5</td><td>35</td></tr></table>		Mol. weight	LogP	Rotatable bonds	Aromatic rings	Heavy atoms	A	341.4	5.23	4	4	26	B	463.5	4.43	4	5	35
	Mol. weight	LogP	Rotatable bonds	Aromatic rings	Heavy atoms														
A	341.4	5.23	4	4	26														
B	463.5	4.43	4	5	35														
Molecular similarity																			
2D similarity																			
3D similarity																			
Biological similarity	<table><tr><td></td><td>Vascular endothelial growth factor receptor 2</td><td>Tyrosine-protein kinase TIE-2</td></tr><tr><td>A</td><td>active</td><td>inactive</td></tr><tr><td>B</td><td>active</td><td>active</td></tr></table>		Vascular endothelial growth factor receptor 2	Tyrosine-protein kinase TIE-2	A	active	inactive	B	active	active									
	Vascular endothelial growth factor receptor 2	Tyrosine-protein kinase TIE-2																	
A	active	inactive																	
B	active	active																	
Global similarity																			
Local similarity																			

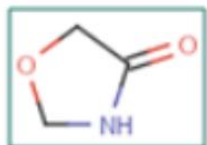
Molecular substructure fingerprint

Representation of a molecular substructure fingerprint with a substructure fingerprint dictionary of given substructure patterns. This molecule is represented in a series of binary bits that represent the presence or absence of substructures in the molecules.

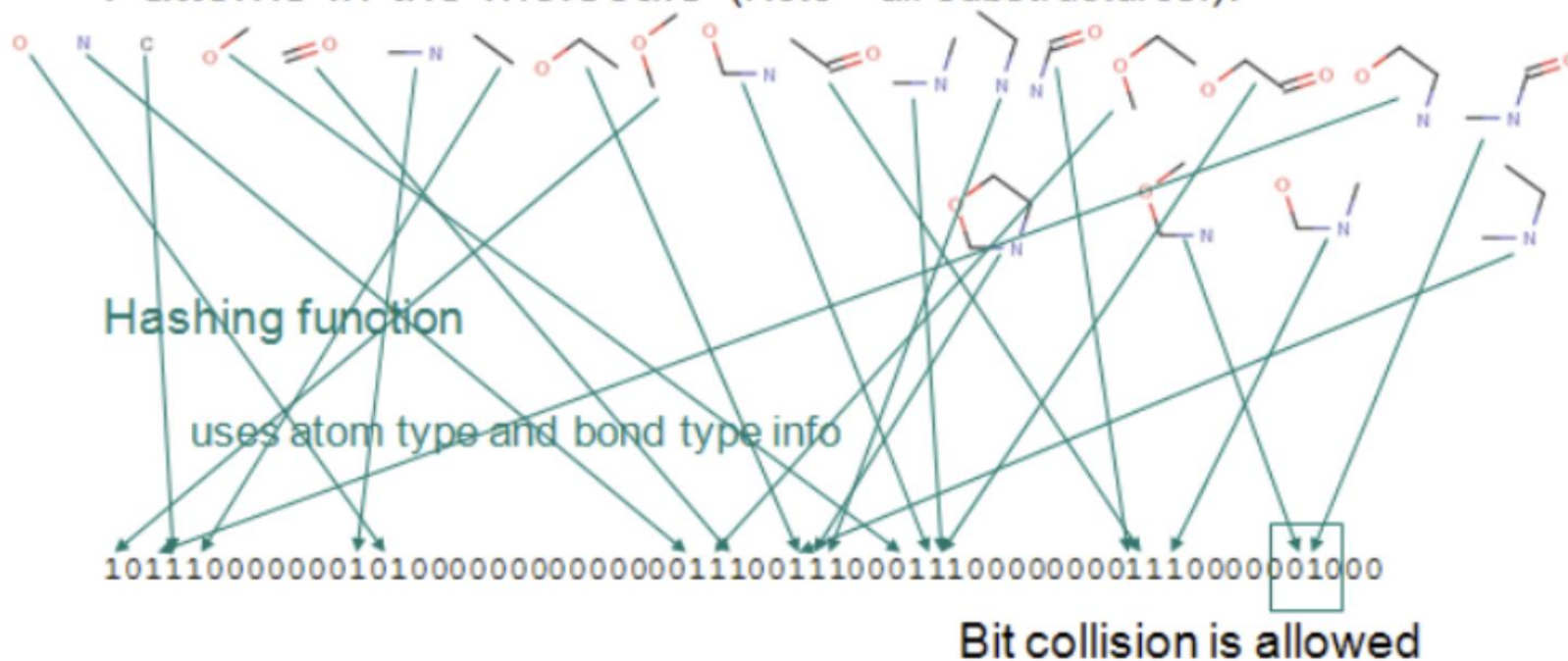


Fingerprint bit-vector: 100000000100000000010

Hashed fingerprint



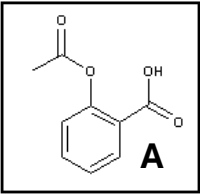
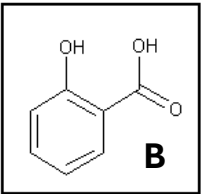
Patterns in the molecule (Note – all substructures!):



Common fingerprints

Fingerprint	Type / Representation	Typical Bit Length	Key Features	Notes / Use Cases
MACCS Keys	Structural key-based	166 bits	Fixed dictionary of predefined substructures (e.g., halogens, rings, bonds)	Simple and interpretable; limited coverage of chemical space
PubChem Fingerprint	Structural key-based	881 bits	Encodes presence of predefined SMARTS patterns	Used in PubChem database; good for database-wide similarity
Morgan (ECFP)	Circular (topological)	Variable (typically 1024 or 2048 bits)	Encodes atom environments within a given radius (e.g., radius=2 → ECFP4)	Widely used in ML; captures substructure diversity effectively
RDKit Fingerprint	Path-based	Variable (1024–2048 bits typical)	Hashes atom–bond paths up to a certain length	Fast and general-purpose; good for quick similarity screening
Atom Pair Fingerprint	Distance-based (topological)	Variable (2048+ bits typical)	Encodes atom types and topological distances	Sensitive to molecular shape; larger bit vectors
Topological Torsion Fingerprint	Path-based (torsions)	Variable	Encodes sequences of four bonded atoms (torsions)	Good descriptor of 3D-related connectivity patterns
Avalon Fingerprint	Hybrid (pattern + substructure)	512–2048 bits	Combines structural keys and path-based features	Compatible with RDKit; flexible for custom workflows
Pharmacophore Fingerprints (e.g., FCFP)	Feature-based circular	Variable	Encodes pharmacophoric features (H-bond donor, acceptor, aromatic, etc.)	Useful in ligand-based screening and QSAR models

Similarity with Tanimoto coefficient

	 A				 B			
A	1	1	0	1	1	0	1	0
B	1	1	0	1	0	0	0	0
								

$$\text{T.c.} = \frac{3}{5+3-3} = 0.6$$

$$\text{Tanimoto coefficient} = \frac{N_{AB}}{N_A + N_B - N_{AB}}$$

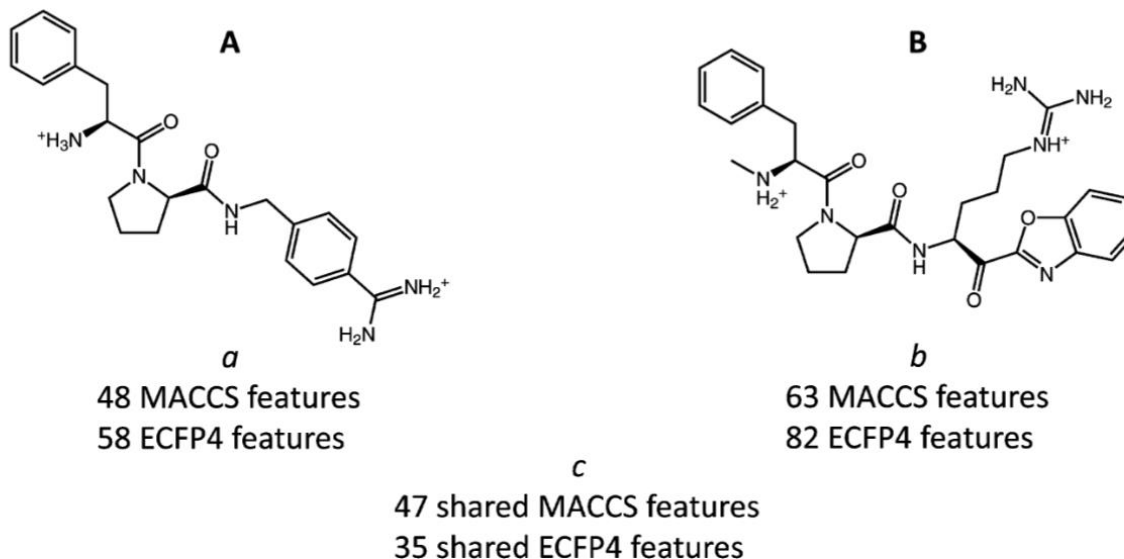
N_A – number of features of A

N_B – number of features of B

C_{AB} – common features of A and B

Two molecules may display a T.c of 1 and not be identical ! (different isomers, tautomers, etc...)

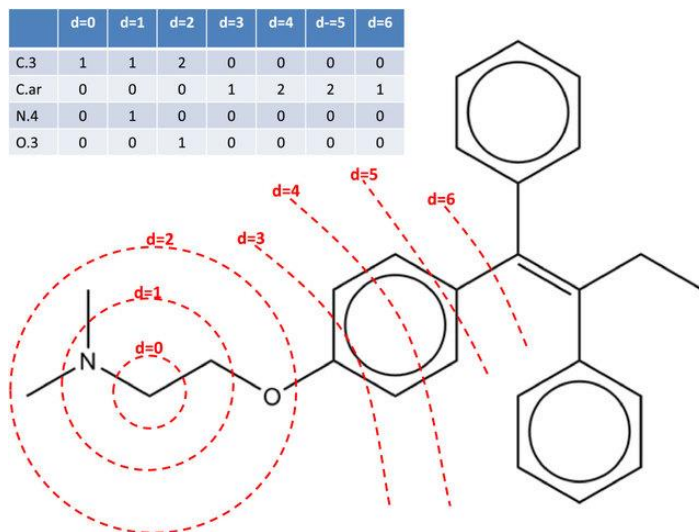
Different measures produce different results



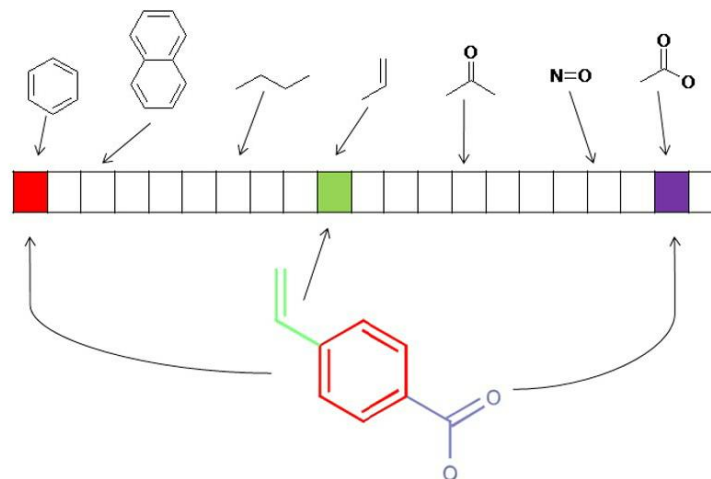
	$Tc(A, B) = \frac{c}{a + b - c}$	$Dc(A, B) = \frac{2c}{a + b}$	$Tv(A, B, \alpha) = \frac{c}{\alpha a + (1 - \alpha)b}$	
			$\alpha = 0.1$	$\alpha = 0.9$
MACCS	0.73	0.85	0.76	0.95
ECFP4	0.33	0.50	0.44	0.58

Comparison of similarity coefficients. For two thrombin inhibitors Dice, Tanimoto, and Tversky coefficients are compared using MACCS and ECFP4. Tversky similarity calculations were carried out using different parameter settings.

Topological versus circular fingerprints



Circular fingerprints encode atomic environments around specific atoms.
(ex: Morgan)



Linear fingerprints encode functional groups in structure.

MACCS fingerprint SMARTS representation 1/3

Bit	Pattern	Description
1	?	ISOTOPE
2	[#104]	limit the above def'n since the RDK
3	[#32,#33,#34,#50,#51,#52,#82,#	Group IVa,Va,Vla Rows 4-6
4	[Ac,Th,Pa,U,Np,Pu,Am,Cm,Bk,Cf	actinide
5	[Sc,Ti,Y,Zr,Hf]	Group IIIB,IVB (Sc...)
6	[La,Ce,Pr,Nd,Pm,Sm,Eu,Gd,Tb,D	Lanthanide
7	[V,Cr,Mn,Nb,Mo,Tc,Ta,W,Re]	Group VB,VIB,VII B
8	[!#6;!#1]1~*~*~*~1	QAAA@1
9	[Fe,Co,Ni,Ru,Rh,Pd,Os,Ir,Pt]	Group VIII (Fe...)
10	[Be,Mg,Ca,Sr,Ba,Ra]	Group IIa (Alkaline earth)
11	*1~*~*~*~1	4M Ring
12	[Cu,Zn,Ag,Cd,Au,Hg]	Group IB,IIB (Cu..)
13	[#8]~[#7](~[#6])~[#6]	ON(C)C
14	[#16]-[#16]	S-S
15	[#8]~[#6](~[#8])~[#8]	OC(O)O
16	[!#6;!#1]1~*~*~*~1	QAA@1
17	[#6]#[#6]	CTC
18	[#5,#13,#31,#49,#81]	Group IIIA (B...)
19	*1~*~*~*~*~*~*~1	7M Ring
20	[#14]	Si
21	[#6]=[#6](~[#6;!#1])~[#6;!#1]	C=C(Q)Q
22	*1~*~*~*~1	3M Ring
23	[#7]~[#6](~[#8])~[#8]	NC(O)O
24	[#7]-[#8]	N-O

Bit	Pattern	Description
25	[#7]~[#6](~[#7])~[#7]	NC(N)N
26	[#6]=;@[#6](@*)@*	C\$=C(\$A)\$A
27	[I]	I
28	[!#6;!#1]~[CH2]~[!#6;!#1]	QCH2Q
29	[#15]	P
30	[#6]~[!#6;!#1](~[#6])(~[#6])~*	CQ(C)(C)A
31	[!#6;!#1]~[F,Cl,Br,I]	QX
32	[#6]~[#16]~[#7]	CSN
33	[#7]~[#16]	NS
34	[CH2]=*	CH2=A
35	[Li,Na,K,Rb,Cs,Fr]	Group IA (Alkali Metal)
36	[#16R]	S Heterocycle
37	[#7]~[#6](~[#8])~[#7]	NC(O)N
38	[#7]~[#6](~[#6])~[#7]	NC(C)N
39	[#8]~[#16](~[#8])~[#8]	OS(O)O
40	[#16]-[#8]	S-O
41	[#6]#[#7]	CTN
42	F	F
43	[!#6;!#1;!H0]~*~[!#6;!#1;!H0]	QHAQH
44	[!#1;!#6;!#7;!#8;!#9;!#14;!#15;!#	OTHER
45	[#6]=[#6]~[#7]	C=CN
46	Br	BR
47	[#16]~*~[#7]	SAN
48	[!#6;!#1](~[#8])(~[#8])	OQ(O)O

MACCS fingerprint SMARTS representation 2/3

Bit	Pattern	Description
72	[#8]~*~*~[#8]	OAAO
73	[#16]=*	S=A
74	[CH3]~*~[CH3]	CH3ACH3
75	*!@[#7]@*	A!N\$A
76	[#6]=[#6](~*)~*	C=C(A)A
77	[#7]~*~[#7]	NAN
78	[#6]=[#7]	C=N
79	[#7]~*~*~[#7]	NAAN
80	[#7]~*~*~*~[#7]	NAAAN
81	[#16]~*(~*)~*	SA(A)A
82	*~[CH2]~[!#6;!#1;!H0]	ACH2QH
83	[!#6;!#1]1~*~*~*~*~1	QAAAA@1
84	[NH2]	NH2
85	[#6]~[#7](~[#6])~[#6]	CN(C)C
86	[C;H2,H3][!#6;!#1][C;H2,H3]	CH2QCH2
87	[F,Cl,Br,I]!.*@*	X!A\$A
88	[#16]	S
89	[#8]~*~*~*~[#8]	OAAAO 90: ('[\$([!#6;!#1;!H0]~*~*~
92	[#8]~[#6](~[#7])~[#6]	OC(N)C
93	[!#6;!#1]~[CH3]	QCH3
94	[!#6;!#1]~[#7]	QN
95	[#7]~*~*~[#8]	NAAO
96	*1~*~*~*~*~1	5 M ring
97	[#7]~*~*~*~[#8]	NAAAO

Bit	Pattern	Description
98	[!#6;!#1]1~*~*~*~*~*~1	QAAAAA@1
99	[#6]=[#6]	C=C
100	*~[CH2]~[#7]	ACH2N 101: ('[\$([R]@1@[R]@[R]@
102	[!#6;!#1]~[#8]	QO
103	Cl	CL
104	[!#6;!#1;!H0]~*~[CH2]~*	QHACH2A
105	*@*(~*)@*	A\$A(\$A)\$A
106	[!#6;!#1]~*~([!#6;!#1])~[!#6;!#1]	QA(Q)Q
107	[F,Cl,Br,I]~*(~*)~*	XA(A)A
108	[CH3]~*~*~*~*~[CH2]~*	CH3AAACH2A
109	*~[CH2]~[#8]	ACH2O
110	[#7]~[#6]~[#8]	NCO
111	[#7]~*~[CH2]~*	NACH2A
112	*~*(~*)(~*)~*	AA(A)(A)A
113	[#8]!.*.*	Onot%A%A
114	[CH3]~[CH2]~*	CH3CH2A
115	[CH3]~*~[CH2]~*	CH3ACH2A
116	Complex Pattern	CH3AACH2A
117	[#7]~*~[#8]	NAO
118	Complex Pattern	ACH2CH2A > 1
119	[#7]=*	N=A
120	[!#6;R]	Heterocyclic atom > 1 (&...) Spec In
121	[#7;R]	N Heterocycle
122	*~[#7](~*)~*	AN(A)A

MACCS fingerprint SMARTS representation 3/3

Bit	Pattern	Description
123	[#8]~[#6]~[#8]	OCO
124	[!#6;!#1]~[#6;!#1]	QQ
125	?	Aromatic Ring > 1
126	*!@[#8]!@*	A!O!A
127	*@*!@[#8]	A\$A!O > 1 (&...) Spec Incomplete 1
129	Complex Pattern	ACH2AACH2A
130	[!#6;!#1]~[#6;!#1]	QQ > 1 (&...) Spec Incomplete
131	[!#6;!#1;!H0]	QH > 1
132	[#8]~*~[CH2]~*	OACH2A
133	*@*!@[#7]	A\$A!N
134	[F,Cl,Br,I]	X (HALOGEN)
135	[#7]!:*:~*	Nnot%A%A
136	[#8]=*	O=A>1
137	[!C;!c;R]	Heterocycle
138	[!#6;!#1]~[CH2]~*	QCH2A>1 (&...) Spec Incomplete
139	[O;!H0]	OH
140	[#8]	O > 3 (&...) Spec Incomplete
141	[C;H3,H4]	CH3 > 2 (&...) Spec Incomplete
142	[#7]	N > 1
143	*@*!@[#8]	A\$A!O
144	*!*:~*:~*	Anot%A%Anot%A
145	*1~*~*~*~*~*~*1	6M ring > 1
146	[#8]	O > 2
147	Complex Pattern	ACH2CH2A

Bit	Pattern	Description
148	*~[!#6;!#1](~*)~*	AQ(A)A
149	[C;H3,H4]	CH3 > 1
150	*!@*~*!@*	A!A\$A!A
151	[#7;!H0]	NH
152	[#8]~[#6](~[#6])~[#6]	OC(C)C
153	[!#6;!#1]~[CH2]~*	QCH2A
154	[#6]=[#8]	C=O
155	*!@[CH2]!@*	A!CH2!A
156	[#7]~*(~*)~*	NA(A)A
157	[#6]-[#8]	C-O
158	[#6]-[#7]	C-N
159	[#8]	O>1
160	[C;H3,H4]	CH3
161	[#7]	N
162	a	Aromatic
163	*1~*~*~*~*~*~*1	6M Ring
164	[#8]	O
165	[R]	Ring
166	?	Fragments FIX: this can't be done i

SMARTS and SMIRKS

- **SMILES** – compact string-based representation for the structure of a molecule
- **SMARTS** – **SMILES ARbitrary Target Specification**
 - An **extension of SMILES** designed for **molecular pattern matching**. It doesn't describe a specific molecule, but a *family* of molecular features or substructures.
 - To be used in chemical databases to find all molecules that contain a specific functional group or structural motif. It's also crucial in cheminformatics tools for defining chemical rules.
- **SMIRKS** - **SMiles Reaktion Specification**
 - A combination of SMILES and SMARTS designed to describe chemical reactions.
 - To encode chemical transformations for reaction modeling, automated reaction enumeration, and force field parameterization

SMARTS / SMIRKS syntax reference

Atomic Specifications:

- [#N] = Atom with atomic number N (e.g., [#6] = carbon, [#7] = nitrogen, [#8] = oxygen)
- [C] or C = Aliphatic carbon
- [c] or c = Aromatic carbon
- [CH3], [CH2], etc. = Carbon with specific number of hydrogens
- [C;H3,H4] = Carbon with 3 or 4 hydrogens
- [!#6] = Not carbon (! means NOT)
- [!#6;!#1] = Neither carbon nor hydrogen
- [!H0] = Has at least one hydrogen
- [R] = In a ring
- [!+0] = Charged atom (not charge 0)
- * = Any atom (wildcard)
- a = Any aromatic atom

Bond Types:

- ~ = Any bond (single, double, triple, or aromatic)
- - = Single bond
- = = Double bond
- # = Triple bond
- : = Aromatic bond
- @ = Ring bond
- !@ = Non-ring bond
- !: = Non-aromatic bond
- =;@ = Double bond in a ring

Ring Patterns:

- *1~*~*~1 = 3-membered ring
- *1~*~*~*~1 = 4-membered ring
- *1~*~*~*~*~1 = 5-membered ring
- *1~*~*~*~*~*~1 = 6-membered ring
- And so on for larger rings

Logical Operators:

- ; = AND (both conditions must be true)
- , = OR (either condition must be true, used within \$(...))
- ! = NOT (negation)

Recursive SMARTS:

- \$(...) = Recursive pattern - allows grouping of alternative patterns
- \$([pattern1],[pattern2]) = Match pattern1 OR pattern2

Branching:

- [#6](~[#6]) = Carbon bonded to another carbon
- [#6](~[#6])(~[#6])~* = Carbon bonded to two carbons and any other atom

Examples from the table:

- [#8]~[#6](~[#8])~[#8] = Oxygen bonded to carbon, which is also bonded to two other oxygens (carbonate-like)
- *@*!@*@* = Ring atom - non-ring bond - atom - ring bond - ring atom (ring-chain-ring pattern)
- c:n = Aromatic carbon bonded to aromatic nitrogen
- \$([CH3]~*~*~[CH2]~*), \$([CH3]~*1~*~[CH2]1) =
• Either a linear CH3-A-A-CH2-A pattern OR a ring pattern with CH3 and CH2

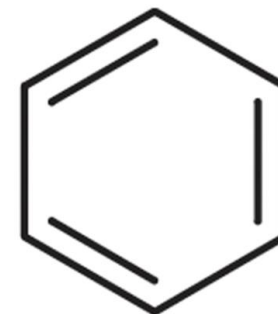
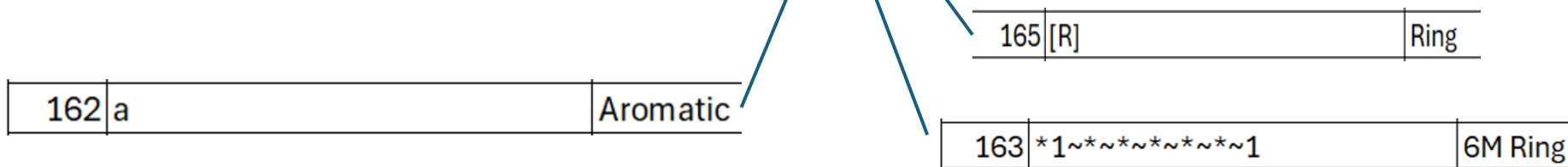
SMARTS examples

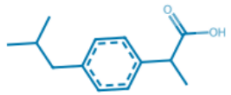
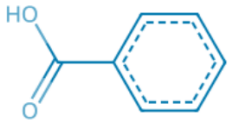
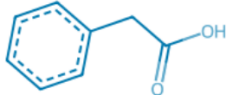
SMARTS Pattern	Description	Example Molecules It Matches
<chem>[OH]</chem>	Any hydroxyl group	Ethanol, water, phenol
<chem>[CX3]=[OX1]</chem>	Carbonyl group (C=O)	Acetone, formaldehyde, amides
<chem>[#6]</chem>	Any carbon atom	Every organic molecule
<chem>[c]</chem>	Any aromatic carbon	Benzene, pyridine, naphthalene
<chem>[N;!H0]</chem>	Nitrogen with at least one H (H-bond donor)	Aniline, ammonium ions, primary amines
<chem>[O;H1]</chem>	Oxygen with exactly one H (e.g., alcohol)	Methanol, serine (side chain)
<chem>[\$(C=O)]</chem>	Carbon atom in a carbonyl (recursive syntax)	Acetaldehyde, acetamide, acetic acid
<chem>[#7]</chem>	Any nitrogen atom	Ammonia, aniline, pyridine, nitriles
<chem>[!C;!c]</chem>	Any atom that is not carbon	Oxygen in water, nitrogen in ammonia
<chem>[R]</chem>	Any atom in any ring	Atoms in cyclohexane, benzene, camphor

MACCS benzene fingerprint

```
# Generate MACCS key
mol = Chem.MolFromSmiles("c1ccccc1")
fp = MACCSkeys.GenMACCSKeys(mol)

print(fp.GetNumBits())          # 167 bits (bit 0 unused)
print(fill(fp.ToBitString(),60))
```

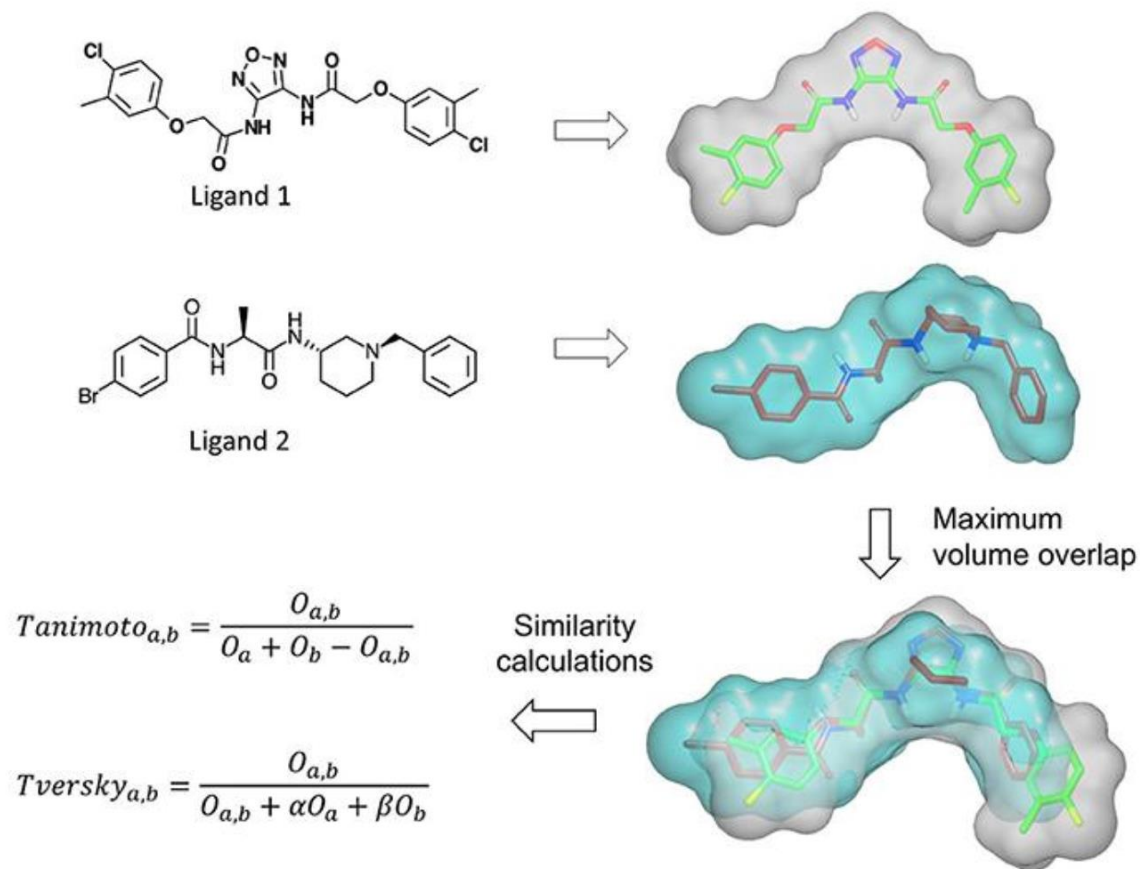
[illegible]

Top Similar Molecules to Ibuprofen (Sorted by Tanimoto Similarity)				
Rank	Name	CID	Tanimoto Similarity	Structure
1	Ibuprofen	3672	1.0000	
2	Benzoic acid	243	0.6000	
3	2-phenyl acetic acid	999	0.5882	
4	(E)-3-phenylprop-2-enoic acid	444539	0.5625	

Searching PubChem for molecules like Ibuprofen, using a MACCS fingerprint and the Tanimoto similarity index.

(Ibuprofen finds itself on the first entry, with Tanimoto similarity of 1.0)

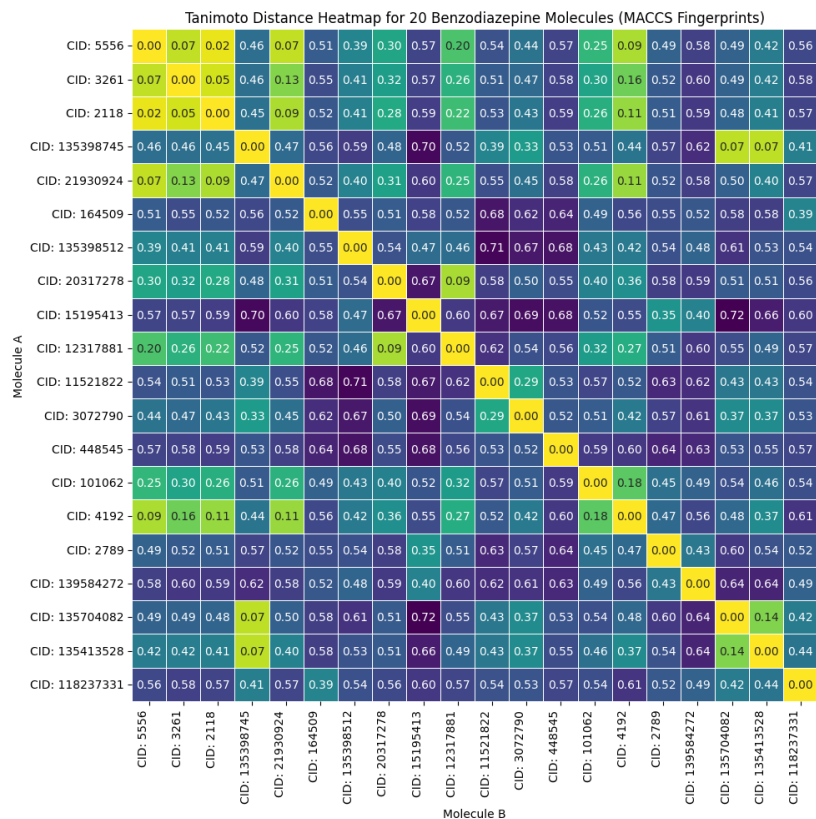
3D similarity (molecular volume overlap)



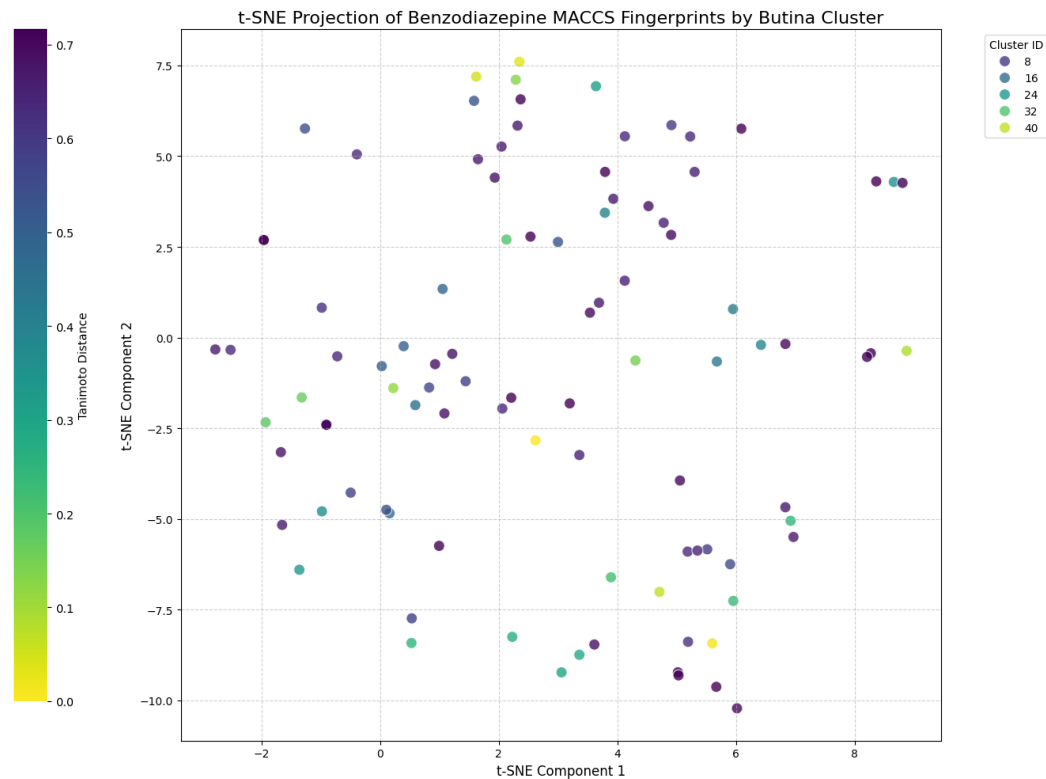


A similarity search for benzodiazepines in the ChEMBL database returned 100 hits.

Searching PubChem for benzodiazepinas

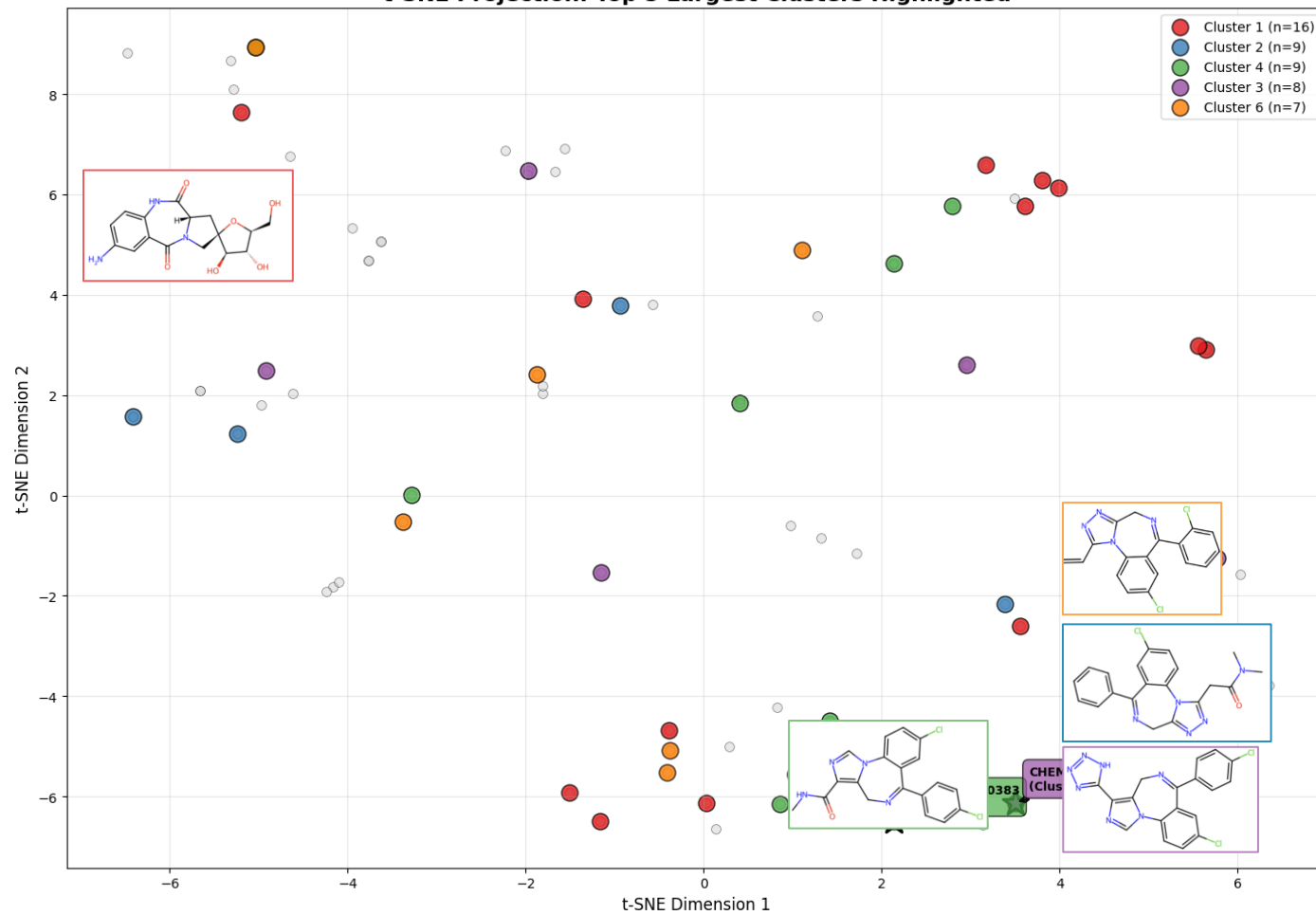


Tanimoto distances between all MACCS fingerprint key pairs are calculated



Molecules are clustered according to similarity (colours denote different clusters).

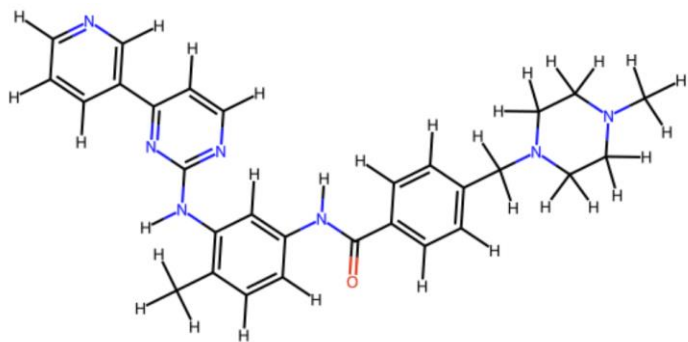
t-SNE Projection: Top 5 Largest Clusters Highlighted



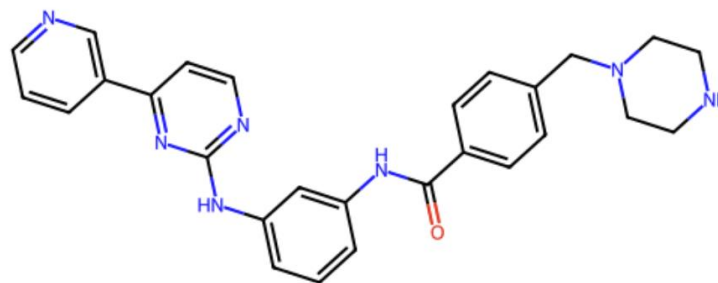
Molecular scaffolds

- **Core frameworks of bioactive molecules;** scaffolds represent the stable backbone or central structure common to a family of compounds
- **Simplify chemical diversity:** by focusing on the core, scaffolds help categorize and compare large libraries of molecules.
- **Central to medicinal chemistry;** scaffold hopping and optimization drive discovery of new leads with similar activity but novel structures.
- **Extracted computationally:** using algorithms like *Bemis–Murcko decomposition* to strip side chains and retain ring systems and linkers.
- **Useful for visualization and design;** scaffold trees and networks reveal structure–activity relationships and chemical space organization.

Bemis-Murcko scaffold for Imatinib

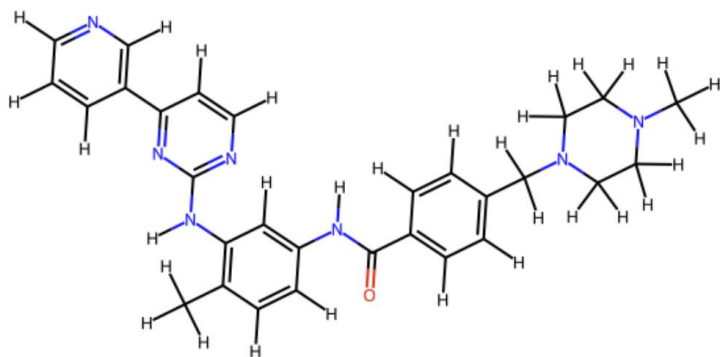


Imatinib

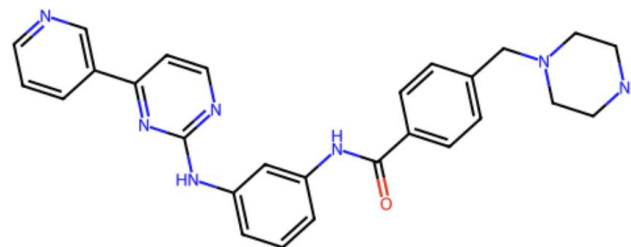


Bemis-Murcko scaffold

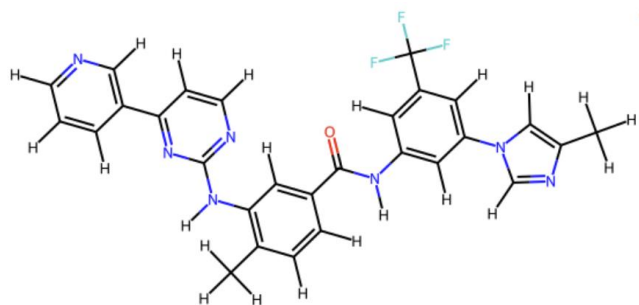
Scaffold hopping



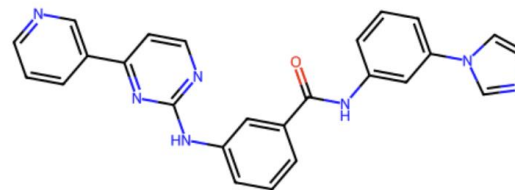
Imatinib



Bemis-Murcko scaffold



Nilotinib



Bemis-Murcko scaffold

Distinct scaffolds, same activity – scaffold hopping.

